prolate spheroids, 141
protons, 22, 24, 27, 49, 74, 142
pulsars, 141–43, 153, 186–87

quantized energy, 19
quantum gravity, 18
quantum mechanics, 18–19, 112
quantum physics, 87
quark-antiquark pairs, 25
quarks, 21–22, 23–25, 27
  attractive forces between, 22–23
  charge of, 22
  name origin of, 22
  subspecies of, 21–22
quasars, 70–71
  lensing of, 72

radio astronomy, 157
radioisotope thermoelectric generators (RTGs), 132
radio telescopes, 155–60, 163, 185, 186
radio waves, 51, 151, 153, 155, 156–57, 163, 184–85, 186–87
Reber, Grote, 157–58
recession, 105
red giants, 139
red stars:
  spectra of, 123
  technetium in, 125–26
relativity, general theory of, 18, 19, 44, 52, 72, 96, 98–102, 104, 111
resolution, 155
Riess, Adam, 103
Roche, Édouard, 138
Roche lobe, 138–39
rogue planets, 82
rotation rates, 141–43
rubies, 123
Rubin, Vera, 80–81
runaway stars, 64

Sagan, Carl, 181, 197*n*
Sagittarius Dwarf, 66
salt, 117, 121
satellites, 175
Saturday Night Live, 39–40
Saturn, 123, 129, 141, 171, 174
Schmidt, Brian, 103, 104
scientific models, 98–99
scintillator, 161
selenium, 127–28
September 11, 2001 terrorist attacks, 179
Shakespeare, William, 173
Shoemaker, Carolyn, 177
Shoemaker-Levy 9 (comet), 42
silicon, 121
silver chloride, 150
single-celled organisms, 150